

例：用直接三角分解法求解方程组

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 3 & 1 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 14 \\ 18 \\ 20 \end{pmatrix}$$

解：

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 3 & 1 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix} = LU$$

计算 U 的第一行：

$$u_{11} = 1, \quad u_{12} = 2, \quad u_{13} = 3$$

计算 L 的第一列：

$$a_{21} = l_{21} u_{11} = 2, \quad \therefore l_{21} = 2$$

$$a_{31} = l_{31} u_{11} = 3, \quad \therefore l_{31} = 3$$

计算 U 的第二行：

$$a_{22} = l_{21} u_{12} + u_{22} = 5$$

$$\therefore u_{22} = 5 - 2 \times 2 = 1$$

$$a_{23} = l_{21} u_{13} + u_{23} = 2$$

$$\therefore u_{23} = 2 - 2 \times 3 = -4$$

计算 L 的第二列：

$$a_{32} = l_{31} u_{12} + l_{32} u_{22} = 1$$

$$\therefore l_{32} = (a_{32} - l_{31} u_{12}) / u_{22} = (1 - 3 \times 2) / 1 = -5$$

计算 U 的第三行：

$$a_{33} = l_{31} u_{13} + l_{32} u_{23} + u_{33} = 5$$

$$\therefore u_{33} = a_{33} - l_{31} u_{13} - l_{32} u_{23} = 5 - 3 \times 3 - (-5) \times (-4) = -24$$

$$A = LU = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -5 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -4 \\ 0 & 0 & -24 \end{pmatrix}$$

$$\therefore Ly = b$$

$$\therefore \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -5 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 14 \\ 18 \\ 20 \end{pmatrix}$$

$$\therefore \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 14 \\ -10 \\ -72 \end{pmatrix}$$

$$\therefore Ux = y$$

$$\therefore \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -4 \\ 0 & 0 & -24 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 14 \\ -10 \\ -72 \end{pmatrix}$$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$